



Efficacy and safety of ustekinumab, an IL-12 and IL-23 inhibitor, in patients with active systemic lupus erythematosus: results of a multicentre, double-blind, phase 2, randomised, controlled study

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Summary

Background Ustekinumab is a monoclonal antibody targeting interleukin (IL)-12 and IL-23 and is approved for the treatment of plaque psoriasis, psoriatic arthritis, and Crohn's disease. IL-12 and IL-23 have been implicated in systemic lupus erythematosus. We aimed to assess the efficacy and safety of ustekinumab for the treatment of systemic lupus erythematosus in patients with moderate-to-severe disease activity despite conventional treatment.

Methods This was a multicentre, double-blind, phase 2, randomised, controlled trial of adult patients with active, seropositive systemic lupus erythematosus, done at 44 private practices and academic centres in Argentina, Australia, Germany, Hungary, Mexico, Poland, Spain, Taiwan, and the USA. Eligible adults were aged 18–75 years, weighed at least 35 kg, and had a diagnosis of systemic lupus erythematosus at least 3 months before the first administration of study drug. Eligible patients were randomly assigned (3:2) to the ustekinumab or placebo group using an interactive web response system with stratification by skin biopsy, lupus nephritis presence, baseline systemic lupus erythematosus medications and systemic lupus erythematosus disease activity index 2000 (SLEDAI-2K) score combined factor, site, region, and race. Patients and investigators were masked to treatment allocation. Patients received an intravenous infusion of ustekinumab (260 mg for patients weighing 35–55 kg, 390 mg for patients weighing >55 kg and ≤85 kg, and 520 mg for patients weighing >85 kg) followed by subcutaneous injections of ustekinumab 90 mg every 8 weeks or intravenous infusion of placebo at week 0 followed by subcutaneous injections of placebo every 8 weeks, both in addition to standard-of-care therapy. The primary endpoint was the proportion of patients achieving a SLEDAI-2K responder index-4 (SRI-4) response at week 24. Efficacy analyses were done in a modified intention-to-treat population of patients who received at least one dose (partial or complete, intravenous or subcutaneous) of their randomly assigned study treatment. Safety analyses were done in all patients who received at least one dose of study treatment, regardless of group assignment. This study is registered at ClinicalTrials.gov, number NCT02349061.

Findings Between Oct 6, 2015, and Nov 30, 2016, 166 patients were screened, of whom 102 were randomly assigned to receive ustekinumab (n=60) or placebo (n=42). At week 24, 37 (62%) of 60 patients in the ustekinumab group and 14 (33%) of 42 patients in the placebo group achieved an SRI-4 response (percentage difference 28% [95% CI 10–47], p=0·006). Between week 0 and week 24, 47 (78%) of 60 patients in the ustekinumab group and 28 (67%) of 42 patients in the placebo group had at least one adverse event. Infections were the most common type of adverse event (27 [45%] in the ustekinumab group vs 21 [50%] in the placebo group). No deaths or treatment-emergent opportunistic infections, herpes zoster, tuberculosis, or malignancies occurred between weeks 0–24.

Interpretation The addition of ustekinumab to standard-of-care treatment resulted in better efficacy in clinical and laboratory parameters than placebo in the treatment of active systemic lupus erythematosus and had a safety profile consistent with ustekinumab therapy in other diseases. The results of this study support further development of ustekinumab as a novel treatment in systemic lupus erythematosus.

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Introduction

Systemic lupus erythematosus is a disease of immune dysregulation characterised by protean clinical manifestations that can affect many organ systems and can result in progressive organ damage. Its pathogenesis is

characterised by the activation of various inflammatory pathways, often in the presence of autoantibodies. Numerous immunological abnormalities have been described, but which of these are central or initiating events and which are consequences of inflammation remains

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Research in context

Evidence before this study

We searched PubMed on May 29, 2018, for reports published in English up to May 29, 2018, using the search terms “ustekinumab”, “lupus”, “systemic lupus erythematosus”, “SLE”, “anti-interleukin-12”, “anti-interleukin-23”, and “anti-interleukin-17”. We found no studies of the efficacy of ustekinumab or other anti-interleukin-12/23 (IL-12/23) p40 drugs, anti-IL-23p19, or anti-IL-17 or anti-IL-17 receptor agents in patients with systemic lupus erythematosus. Published literature implicates both the IL-12 and IL-23 pathways in the pathogenesis of systemic lupus erythematosus. Ustekinumab is a monoclonal anti-IL-12 and IL-23 antibody that is approved for use in adults with psoriatic arthritis and Crohn’s disease and adult and adolescent patients with plaque psoriasis.

Added value of this study

To the best of our knowledge, this is the first phase 2, randomised, placebo-controlled study of ustekinumab in patients with active systemic lupus erythematosus. Patients who received ustekinumab had greater improvements in clinical parameters

(global systemic lupus erythematosus disease activity responder index, joint counts, mucocutaneous disease, and disease flares) and laboratory parameters (C3 complement concentrations and anti-dsDNA autoantibodies) at week 24 than those who received placebo. Safety results were consistent with the known safety profile of ustekinumab. These results strongly support the further investigation of ustekinumab as a novel mechanism of action for patients with systemic lupus erythematosus.

Implications of all the available evidence

Studies of B-cell-targeted therapies in systemic lupus erythematosus have had mixed results, standard therapies carry considerable toxicity, and only a single drug with a novel mechanism of action has been approved for systemic lupus erythematosus in almost six decades. Thus, a significant unmet need remains for these patients. Results of our study suggest that ustekinumab might be an effective therapy in systemic lupus erythematosus. The efficacy and safety of ustekinumab in patients with systemic lupus erythematosus is being further assessed in a phase 3 programme.

unclear.¹ The study of systemic lupus erythematosus is further challenged by the inherent clinical and biological heterogeneity of the disease.

Conventional therapy for patients with moderate-to-severe systemic lupus erythematosus uses glucocorticoids for rapid control of inflammation and clinical manifestations, and antimalarial medications or immunosuppressive drugs, or both, for longer-term control of the disease.² Belimumab, which targets B-lymphocyte stimulator, is the only monoclonal antibody approved as an add-on therapy for adults with active, autoantibody-positive systemic lupus erythematosus.³ The phase 3 trials of belimumab showed an enriched response in a subset of patients with greater serological and clinical disease activity at baseline.⁴

Advances in the quality of systemic lupus erythematosus care have led to higher overall survival rates in recent years,⁵ improvement in general outcomes,⁶ and better control of some of the more common organ-threatening complications, such as lupus nephritis.⁷ Nevertheless, treatments for systemic lupus erythematosus have not achieved many of the other therapeutic goals that are widely accepted as treatment targets in other rheumatological diseases. For example, durable remission is achieved in only a minority of patients,⁸ the risk for flares remains substantial,⁹ progression to end-stage renal disease is still considerable,¹⁰ work disability is common,¹¹ the use of glucocorticoids can result in a wide spectrum of short-term and long-term toxicities,¹² and it is associated with considerably increased mortality due to cardiovascular disease and other comorbidities, which is believed to result from both the disease and its treatment.¹³ Additionally, health-related quality of life in patients

with systemic lupus erythematosus is often markedly reduced compared with matched healthy controls or patients with other chronic diseases,¹⁴ even in countries where broad access to high-quality health care exists. Thus, there remains a significant unmet need for safe and effective therapies to treat this complex autoimmune disorder.

The development of improved therapies for systemic lupus erythematosus has been very difficult and hampered by uncertainty as to the most suitable molecular or cellular targets for intervention. Although two phase 3 trials with belimumab were successful and led to approval of this therapy, several other trials of interventions targeting B cells and other mechanisms have been ineffective.¹⁵ In a phase 2 study of baricitinib (a Janus kinase inhibitor), the 4-mg dose, but not the 2-mg dose, was effective in patients with active systemic lupus erythematosus.¹⁶ Two successful phase 2 trials targeting the interferon (IFN) type I system have also been published.^{17,18} However, only patients with increased expression of type I IFN-regulated genes (IFN-I signature)¹⁹ at baseline responded significantly better than placebo in a clinical trial targeting the type I IFN receptor.¹⁷ Thus, additional pathways are likely to be important drivers of systemic lupus erythematosus activity.

The cytokines interleukin (IL)-12 and IL-23 are important mediators of immunity that have been implicated in the pathogenesis of systemic lupus erythematosus. IL-12 is essential for T helper 1 (Th1) and T follicular helper cell development and cytotoxic T cell activation and function. IL-23 drives the expansion and survival of pathogenic Th17 and other IL-17-producing cells, which promote inflammation in tissues. Cytokine dysregulation, including increase of IL-12, IL-6, and IL-23 concentrations, has

been reported in patients with systemic lupus erythematosus,^{20,21} suggesting that the inflammatory environment in systemic lupus erythematosus is prone to inducing Th1 or Th17 cells, or both. Genetic risk for developing systemic lupus erythematosus has been associated with the IL-12 and IL-12 receptor pathway and downstream signalling mediators such as tyrosine kinase TYK2 and signal transducer and activator of transcription 4.²² Lessening of disease severity in murine lupus models deficient in IL-12²³ and IL-23^{20,24} also suggests that these cytokines are key elements in the pathogenesis of systemic lupus erythematosus. Additionally, a large meta-analysis that aimed to identify novel targets and potential drugs for systemic lupus erythematosus highlighted ustekinumab as the top scoring candidate for repositioning in systemic lupus erythematosus.²⁵

Ustekinumab is a fully human monoclonal antibody directed at the p40 subunit shared by the cytokines IL-12 and IL-23. It has been approved worldwide for the treatment of psoriatic arthritis and Crohn's disease in adults and plaque psoriasis in adults and adolescents.²⁶

On the basis of research implicating IL-12 and IL-23 in systemic lupus erythematosus and the established safety profile of ustekinumab in other indications, we designed a phase 2 clinical trial to assess the efficacy and safety of ustekinumab for the treatment of systemic lupus erythematosus in patients with moderate-to-severe disease activity despite conventional treatment.

Methods

Study design

This multicentre, double-blind, phase 2, randomised, controlled trial was done at 44 private practices and academic centres in Argentina, Australia, Germany, Hungary, Mexico, Poland, Spain, Taiwan, and the USA. This trial was done in accordance with the principles of the Declaration of Helsinki and Good Clinical Practice. The protocol was approved by the Institutional Review Board or Ethics Committee at each site. The protocol is available in the appendix.

Patients

Eligible adults were aged 18–75 years, weighed at least 35 kg, and had a diagnosis of systemic lupus erythematosus at least 3 months before the first administration of study drug. Systemic lupus erythematosus was defined by the Systemic Lupus International Collaborating Clinics classification criteria.²⁷ Patients had to meet at least four criteria with at least one clinical and one immunological criterion, or have a diagnosis of lupus nephritis in the presence of antinuclear antibodies or anti-double-stranded DNA (dsDNA) antibodies. All patients were required to have tested positive during screening for autoantibodies (antinuclear antibodies titre >1:80, anti-dsDNA antibodies >75 IU/mL, or anti-Smith antibodies >120 arbitrary units per mL), in addition to having a positive titre in their medical history. Patients also had to

have at least one British Isles Lupus Assessment Group 2004 index (BILAG-2004) domain A (severe manifestation) or two BILAG domain B (moderate manifestation) organ domain scores during screening, or both. Eligible patients also had a systemic lupus erythematosus disease activity index 2000 (SLEDAI-2K) score of at least 6 during screening (assessed about 2–6 weeks before group assignment) and a SLEDAI-2K score of at least 4 for clinical features (excluding laboratory results) despite background treatment (eg, immunomodulators, antimalarial drugs, glucocorticoids, non-steroidal anti-inflammatory drugs [NSAIDs], or topical medications, or a combination of these) at baseline before the first administration of the study drug. Patients with inflammatory diseases that might confound efficacy assessments (eg, rheumatoid arthritis, psoriatic arthritis, and psoriasis) were excluded, as were patients with unstable or progressive systemic lupus erythematosus (eg, active class III or IV glomerulonephritis, systemic vasculitis, or active CNS involvement). Other inclusion and exclusion criteria are listed in the appendix.

Concomitant use of specified medications was permitted during the study at stable doses as follows: oral glucocorticoids (≤ 20 mg per day, prednisone or equivalent) for at least 4 weeks before the first administration of study drugs, immunosuppressant drugs (eg, mycophenolate mofetil or mycophenolic acid ≤ 2 g per day, azathioprine or 6-mercaptopurine ≤ 2 mg/kg per day, or methotrexate ≤ 25 mg per week) for at least 6 weeks, antimalarials (eg, hydroxychloroquine, quinacrine, or chloroquine) for at least 8 weeks (with a stable dose for ≥ 6 weeks), angiotensin-converting enzyme inhibitors or angiotensin receptor blockers for at least 2 weeks, or NSAIDs or other analgesics for at least 2 weeks. Adjustments to concomitant medication were generally not permitted, with the following exceptions: an increase or decrease in prednisone (or equivalent) of 5 mg or less per day were permitted between week 0 and week 6; and decreases in prednisone (or equivalent) of 5 mg or less per day was allowed between week 6 and week 12 if the dose was increased between week 0 and week 6. Patients were required to provide written informed consent before any study-related procedures were performed.

Randomisation and masking

Patients were randomly assigned (3:2) to receive either ustekinumab or placebo at baseline. Dynamic central randomisation was performed using an interactive web response system with stratification by the following factors: skin biopsy (yes or no), presence of lupus nephritis (yes or no; in the opinion of the investigator), baseline systemic lupus erythematosus medications and SLEDAI-2K score combined factor (four strata: high-dose medications and SLEDAI-2K ≥ 10 , high-dose medications and SLEDAI-2K <10, medium-dose medications and SLEDAI-2K ≥ 10 , medium-dose medications and SLEDAI-2K <10), site, region, and race (white, black, and

See Online for appendix

other). High-dose medications were defined as any of the following: at least 15 mg per week methotrexate, at least 1.5 mg/kg per day azathioprine or 6-mercaptopurine, at least 1.5 g per day mycophenolate mofetil, or at least 15 mg per day prednisone or equivalent, or any combination of these. Patients receiving baseline medication for systemic lupus erythematosus who did not meet the criteria for the category of high-dose medications were included in the medium-dose medication category.

Randomisation codes were kept within the interactive web response system so patients and investigators remained masked to treatment allocation. The funder was masked to treatment allocation up to the week-24 database lock.

Procedures

Ustekinumab was administered as an intravenous infusion at doses based on weight categories (260 mg for patients weighing 35–55 kg, 390 mg for patients weighing >55 kg and ≤85 kg, and 520 mg for patients weighing >85 kg) at week 0, followed by subcutaneous injections of ustekinumab 90 mg at week 8 and every 8 weeks thereafter up to week 40. Patients randomly assigned to the placebo group received an intravenous placebo infusion at week 0 followed by subcutaneous placebo injections at week 8 and week 16. At week 24, patients in the placebo group crossed over to receive subcutaneous ustekinumab 90 mg every 8 weeks up to week 40. Eligible patients continued open-label ustekinumab treatment up to week 104 as part of an ongoing long-term extension. Here we report data up to the week-24 database lock.

The following clinical efficacy assessments were done at every visit, every 4 weeks: SLEDAI-2K score;²⁸ modified SLEDAI-2K response (≥4 point improvement from baseline in SLEDAI-2K score excluding serological markers of disease activity [C3, C4, and anti-dsDNA antibodies]); physician's global assessment (PGA);²⁹ the BILAG-based combined lupus assessment (BICLA) response,³⁰ which is a composite measure composed of at least one gradation improvement from baseline in all body systems with moderate or severe disease activity on the BILAG-2004 index, no new BILAG domain A score and no more than one new BILAG domain B score, no worsening in the total SLEDAI-2K score from baseline, no worsening (<10 mm increase) in the PGA score from baseline, no treatment failure; SLEDAI-2K responder index-4 (SRI-4) response, which is a composite measure requiring at least a 4-point reduction in SLEDAI-2K score, no worsening (<10 mm increase) in the PGA score from baseline, no new BILAG domain A score, and no more than one new BILAG domain B score;³¹ SRI-5 and SRI-6 responses, defined similarly to the SRI-4 response, but requiring a 5-point or greater reduction (SRI-5) or a 6-point or greater reduction (SRI-6) in SLEDAI-2K; the cutaneous lupus erythematosus disease area and severity index (CLASI);³² and assessment of tender, swollen, and active (defined

	Ustekinumab (n=60)	Placebo (n=42)
Age, years	40.0 (12.0)	42.9 (11.3)
Sex		
Women	58 (97%)	35 (83%)
Men	2 (3%)	7 (17%)
Weight, kg	75.4 (20.6)	74.3 (22.8)
Race		
White	42 (70%)	28 (67%)
Black	4 (7%)	3 (7%)
Asian	8 (13%)	6 (14%)
Other	6 (10%)	5 (12%)
Hispanic ethnicity	20 (33%)	12 (29%)
Disease duration, years	9.7 (8.3)	9.5 (7.2)
SLEDAI-2K	10.6 (3.3)	11.4 (4.5)
≥1 BILAG domain A	27 (45%)	22 (52%)
≥2 BILAG domain B	28 (47%)	16 (38%)
PGA (VAS 0–10 cm)	4.9 (1.6)	4.9 (1.6)
Number of active joints	6.8 (6.3)	7.8 (6.0)
Number of joints with pain	10.5 (7.9)	12.2 (9.2)
Number of joints with swelling	6.7 (6.3)	7.2 (6.0)
≥4 joints with pain and signs of inflammation	39 (65%)	31 (74%)
CLASI activity score	7.2 (6.5)	7.4 (6.4)
CLASI activity score ≥4	37 (62%)	24 (57%)
History of lupus nephritis	9 (15%)	7 (17%)
Anti-nuclear antibodies present	54 (90%)	40 (95%)
Anti-double stranded DNA antibodies present	25 (42%)	18 (43%)
Anti-Smith antibodies present	12 (20%)	10 (24%)
Low C3 concentration	25 (42%)	16 (38%)
Low C4 concentration	16 (27%)	15 (36%)
Type I interferon signature*		
High	37 (63%)	30 (73%)
Low	22 (37%)	11 (27%)
Concomitant immunomodulatory and immunosuppressant medications		
Glucocorticoids	51 (85%)	34 (81%)
Dose, mg per day	9.3 (4.7)	10.5 (5.2)
Antimalarial	44 (73%)	26 (62%)
Mycophenolate mofetil	8 (13%)	7 (17%)
Azathioprine or 6-mercaptopurine	12 (20%)	9 (21%)
Methotrexate	12 (20%)	6 (14%)

Data are presented as n (%) or mean (SD) unless otherwise noted. SLEDAI-2K=systemic lupus erythematosus disease activity index 2000. BILAG=British Isles Lupus Assessment Group. PGA=physician's global assessment. VAS=visual analog scale. CLASI=cutaneous lupus erythematosus disease area and severity index. *Denominators are number of patients included in the microarray analysis: 59 in the ustekinumab group and 41 in the placebo group.

Table 1: Baseline patient demographic and disease characteristics

as having both pain or tenderness and signs of inflammation [eg, swelling or effusion]) joint counts.

Safety was monitored throughout the study as events occurred and at every visit, every 4 weeks. Adverse events and serious adverse events are reported according to the

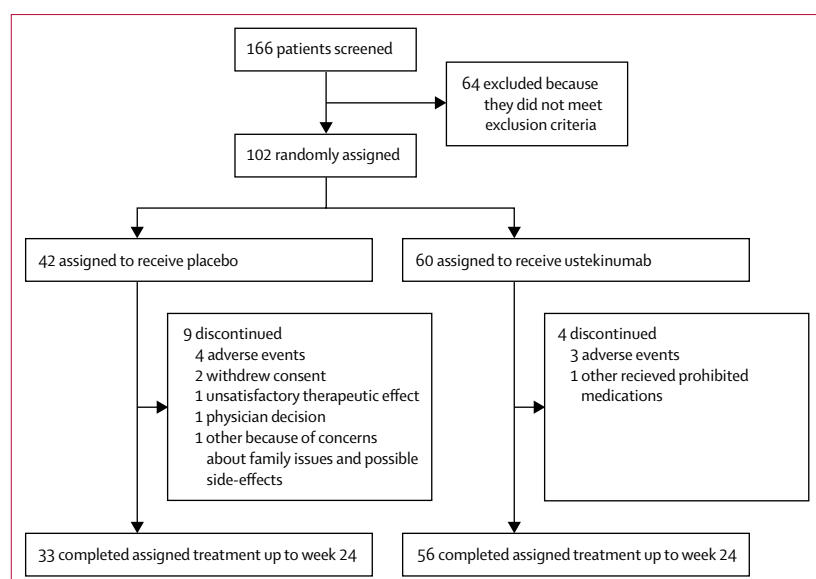


Figure 1: Patient flow diagram

actual treatment received when the event occurred. Serum samples were collected for assessment of ustekinumab concentrations and for the presence of antibodies to ustekinumab, which was analysed using a highly sensitive, drug-tolerant enzyme immunoassay.

We also did a ustekinumab transcriptome microarray analysis (appendix). An IFN-I gene signature composed of 21 genes (appendix) was analysed to establish IFN-I activity in samples from the study population and from age-matched and sex-matched healthy donor samples. For each gene, we calculated the fold change over the median of the healthy donors. To define the score, we calculated the median of the 21-gene fold change for each patient and we segregated the patients with systemic lupus erythematosus into high IFN-I and low IFN-I subsets on the basis of the distribution of the IFN-I gene signature scores using a cutoff of 2.5. Higher scores denote greater IFN-I activity (appendix). For longitudinal analyses, we computed an IFN signature enrichment score for each patient using gene set variation analysis to assess the IFN-I signature score over time.

Outcomes

The primary endpoint was the proportion of patients who achieved an SRI-4 composite response at week 24. Major secondary endpoints, all at week 24, were the change from baseline in SLEDAI-2K, the change from baseline in PGA, and the proportion of patients with BICLA response.

Additional prespecified endpoints included the time to first BILAG flare (defined as at least one new BILAG A score or at least two new BILAG B scores) from week 12 to week 24, proportions of patients with an SRI-5 and SRI-6 response at week 24, proportion of patients with no worsening in BILAG (defined as no new BILAG A score

and no more than one new BILAG B score) at week 24, proportion of patients with no worsening in PGA at week 24, proportion of patients with a SLEDAI-2K response (defined as at least a 4-point improvement from baseline in total SLEDAI-2K score), mean change from baseline in tender, swollen, and active joint counts, and change from baseline serological activity (eg, anti-dsDNA antibodies or C3 or C4 concentrations). The consistency of the primary endpoint was assessed across several subgroups using baseline demographic and disease characteristics. Additional endpoints are in the appendix and will be reported elsewhere.

Post-hoc analyses were done to establish the proportion of patients with a modified SLEDAI-2K response, the proportion of patients with at least a 50% improvement from baseline in the number of active joints (of the patients with at least four involved joints at baseline), the proportion of patients with at least a 50% improvement from baseline in CLASI activity score (of the patients with a baseline CLASI activity score of ≥ 4), the proportion of patients with at least a 30% improvement from baseline in PGA score, and whether the IFN-I signature was modulated after treatment with ustekinumab.

Statistical analysis

Assuming a proportion of SRI-4 responders at week 24 of 35% in the placebo group⁴ and 60% in the ustekinumab group (odds ratio 2.79; difference 25%), 60 patients in the ustekinumab group and 40 patients in the placebo group were projected to yield about 80% power to detect a significant difference using a two-sided Wald test on the odds ratio at an α level of 0.1.

Two interim analyses were done for the primary endpoint to maintain the α level at less than 0.1. The final analysis was adjusted by the α spent from the two interim analyses to a two-sided α level of 0.082. Interim analyses met criteria based on marked efficacy for commencing preparatory work for a potential phase 3 programme. The interim analysis was performed by an unmasked statistical support team and reported to the data monitoring committee as well as a funder representative who was independent from the masked study team. To control for multiplicity, if the primary endpoint was significant as defined, the major secondary endpoints were tested sequentially according to the order specified above. If the previous endpoint was not statistically significant, subsequent endpoints were not tested and nominal *p* values were reported (appendix).

Efficacy analyses were done in a modified intention-to-treat population of patients who received at least one dose (partial or complete, intravenous or subcutaneous) of their randomly assigned study treatment. Safety analyses were done in all patients who received at least one dose of study treatment, regardless of group assignment.

All statistical analyses were performed using SAS, version 9.4 (appendix). On the basis of information provided by clinical trial centres, a masked panel

composed of sponsor personnel and external experts adjudicated all diagnoses of lupus before randomisation and continuously adjudicated scoring of systemic lupus erythematosus assessment instruments throughout the study. An independent data monitoring committee regularly reviewed unmasked interim safety data. This trial was registered with ClinicalTrials.gov, number NCT02349061.

Role of the funding source

Employees of the funder participated in study design, data collection, analysis, and interpretation, and writing of the report. The corresponding author had full access to all the data in the study and all authors drafted or revised the manuscript and gave permission for the final version to be submitted to the journal.

Results

Patients were recruited between Oct 6, 2015, and Nov 30, 2016. Final data collection was on May 15, 2017. 166 patients were screened, of whom 102 were randomly assigned to receive placebo (n=42) or ustekinumab (n=60) in addition to the baseline standard-of-care therapy. Demographic and disease characteristics and concomitant medication use at baseline were well balanced between the treatment groups (table 1). Between week 0 and week 24, 13 patients (nine in the placebo group and four in the ustekinumab group) discontinued study treatment (figure 1). The most common reasons for discontinuing the study drug were adverse events (seven patients) and withdrawal of consent (two patients); one patient in the placebo group discontinued because of unsatisfactory therapeutic effect.

At week 24, a greater proportion of patients in the ustekinumab group achieved the primary endpoint of SRI-4 response than in the placebo group (37 [62%] of 60 patients in the ustekinumab group vs 14 [33%] of 42 patients in the placebo group; percentage difference 28% [95% CI 10–47], $p=0.006$; table 2). The proportions of patients achieving an SRI-4 response were greater in the ustekinumab group than the placebo group at all timepoints between week 12 and week 24 (figure 2A). Prespecified subgroup analyses yielded favourable odds ratios for ustekinumab compared with placebo across a variety of baseline demographic and disease characteristics, suggesting a beneficial effect of ustekinumab in various patient subgroups (appendix).

Patients in the ustekinumab group had a significantly greater mean change from baseline in SLEDAI-2K score at week 24 than those in the placebo group (-4.4 [SD 2.9] in the ustekinumab group vs -3.8 [5.4] in the placebo group; $p=0.093$; table 2). The mean change from baseline in the PGA score was numerically, but not significantly, greater in the ustekinumab group than in the placebo group (table 2). The proportion of patients achieving BICLA response at week 24 was similar between the treatment groups; however, the proportion

	Ustekinumab (n=60)	Placebo (n=42)	Difference (95% CI)	p value
Primary endpoint				
SRI-4 response	37 (62%)	14 (33%)	28% (10 to 47)	0.006
Main secondary endpoints				
Change from baseline in SLEDAI-2K,	-4.4 (2.9)	-3.8 (5.4)	-0.6 (-2.4 to 1.2)	0.093
Change from baseline in PGA	-2.2 (1.9)	-1.9 (2.2)	-0.2 (-1.1 to 0.6)	0.394*
BICLA response	21 (35%)	14 (33%)	2% (-17 to 20)	0.994*
Additional endpoints				
SRI-5 response	26 (43%)	9 (21%)	22% (4 to 40)	0.022*
SRI-6 response	26 (43%)	8 (19%)	24% (7 to 42)	0.012*
No worsening in BILAG score	29 (48%)	11 (26%)	22% (4 to 41)	0.028*
No worsening in PGA†				
Patients, n/N (%)	51/55 (93%)	29/32 (91%)	2% (-9 to 13)	..
Mean response rate, % (95% CI)	92.4% (81.4 to 97.1)	88.9% (73.4 to 95.9)	..	0.312*
SLEDAI-2K response†				
Patients, n/N (%)	38/53 (72%)	15/31 (48%)	23% (4 to 42)	..
Mean response rate, % (95% CI)	76.8% (76.4 to 77.2)	49.1% (48.2 to 50.0)	..	0.007*
Post-hoc endpoints				
Modified SLEDAI-2K response†				
Patients, n/N (%)	40/56 (71%)	18/32 (56%)	15% (-4 to 34)	..
Mean response rate, % (95% CI)	75.0% (61.4 to 85.0)	51.6% (35.4 to 67.4)	..	0.016*
≥50% improvement from baseline joint disease activity†‡				
Patients, n/N (%)	32/37 (86%)	14/23 (61%)	26% (9 to 43)	..
Mean response rate, % (95% CI)	90.1% (75.2 to 96.5)	65.5% (44.6 to 81.7)	..	0.010*
≥50% improvement from baseline CLASI activity score†§				
Patients, n/N (%)	17/32 (53%)	6/17 (35%)	18% (-1 to 37)	..
Mean response rate, % (95% CI)	64.1% (43.0 to 80.9)	29.9% (12.0 to 57.0)	..	0.032*
≥30% improvement from baseline in PGA	37 (62%)	18 (43%)	19% (-1 to 38)	0.082*

Data are n (%) or mean (SD) unless otherwise stated. SRI=SLEDAI-2K responder index. SLEDAI-2K=systemic lupus erythematosus disease activity index 2000. PGA=physician's global assessment. BICLA=BILAG-based combined lupus assessment. BILAG=British Isles Lupus Assessment Group. CLASI=cutaneous lupus erythematosus disease area and severity index. *Nominal p value, not adjusted for multiplicity. †The proportions of patients with responses are the observed values at week 24; the mean response rates are based on multiple imputations for missing data. ‡Patients with at least four joints with pain and signs of inflammation at baseline. §Patients with CLASI activity score of at least 4 at baseline.

Table 2: Efficacy at week 24

of patients with no worsening in BILAG-2004 score was greater in the ustekinumab group than in the placebo group (29 [48%] of 60 patients vs 11 [26%] of 42 patients; $p=0.028$; table 2). Consistent with the primary endpoint at week 24, a greater proportion of patients in the ustekinumab group achieved SRI-5 and SRI-6 responses than those in the placebo group (table 2). The SLEDAI-2K response rate was greater in the ustekinumab group than in the placebo group (76.8% [95% CI 76.4–77.2] vs 49.1% [48.2–50.0]; $p=0.007$), as was the modified SLEDAI-2K response rate (75.0% [61.4–85.0] vs 51.6% [35.4–67.4]; $p=0.016$; post-hoc analysis).

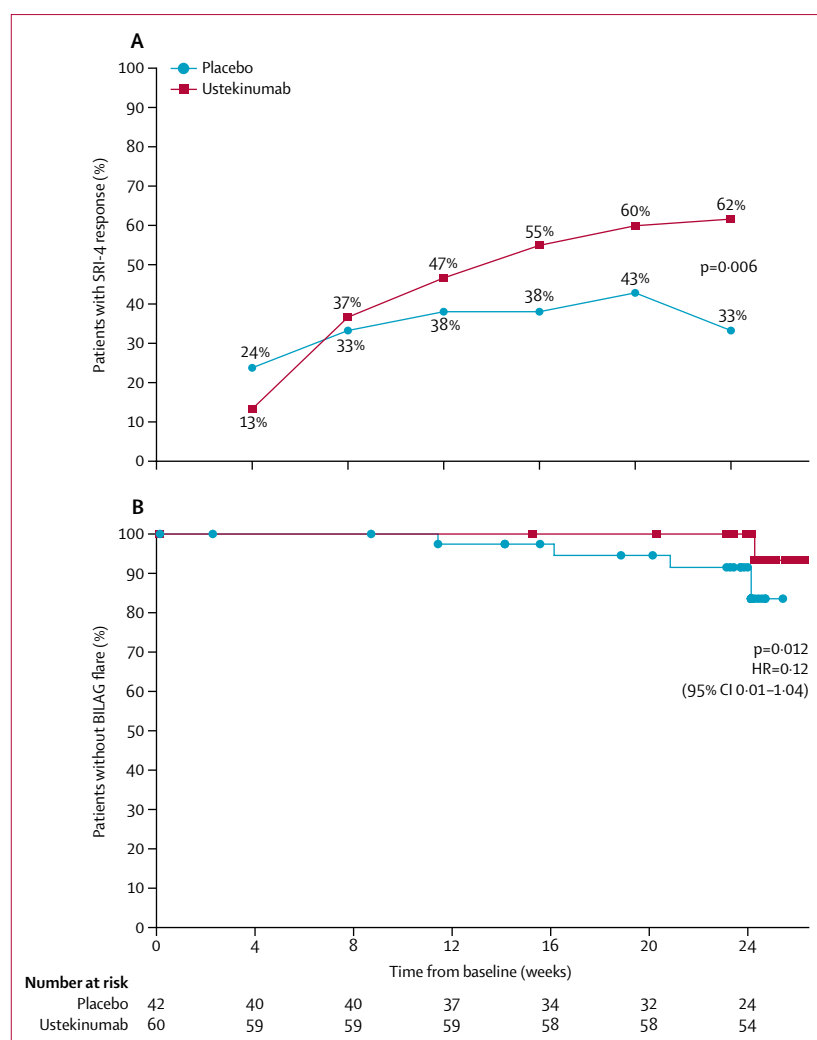


Figure 2: SRI-4 responses, weeks 0–24 (A) and time to BILAG flare, weeks 12–24 (B)

p value calculated by logistic regression with baseline SLEDAI-2K, baseline systemic lupus erythematosus medication use, and race as covariates. Prespecified modified intention-to-treat analysis with treatment failures, dropouts, and missing data considered to be non-responders. SRI-4=SLEDAI-2K responder index-4. SLEDAI-2K=systemic lupus erythematosus disease activity index 2000. BILAG=British Isles Lupus Assessment Group. HR=hazard ratio.

16 patients (seven [17%] of 42 in the placebo group and nine [15%] of 60 in the ustekinumab group) had a history of lupus nephritis, yet clinical evidence of active nephritis (appendix) at baseline was only observed in eight patients (four [10%] and four [7%]). Therefore, the effects of ustekinumab on lupus nephritis could not be adequately assessed in this subpopulation. Assessment of systemic lupus erythematosus disease activity biomarkers showed greater improvements in anti-dsDNA autoantibodies and C3 concentrations in the ustekinumab group than in the placebo group up to week 24; however, changes in C4 concentrations were not consistently different between the treatment groups (appendix).

Improvements in patients in the ustekinumab group compared with those in the placebo group were also

observed across additional measures of disease activity, including assessments of systemic lupus erythematosus flares (figure 2B). Patients who received ustekinumab were less likely to have a BILAG flare between weeks 12 and 24 than those who received placebo (hazard ratio 0.12 [95% CI 0.01–1.04], $p=0.012$; figure 2B). Notably, all observed disease flares were severe (BILAG A).

Mean reductions in the number of active joints at week 24 were not statistically different between the ustekinumab group (-4.5 [SEM 0.6]) and the placebo group (-2.8 [1.3]; $p=0.103$; appendix). Improvement in the mean number of joints with tenderness and swelling as well as only swollen joints, but not in those with only tender joints, was generally numerically greater in patients in the ustekinumab group than those in the placebo group between weeks 12 and 24 (appendix).

In post-hoc analyses at week 24, the ustekinumab group had greater response rates than the placebo group for a 50% minimum improvement from baseline in joint disease activity (90.1% [95% CI 75.2–96.5] vs 65.5% [44.6–81.7]; $p=0.010$) and CLASI activity score (64.1% [43.0–80.9] vs 29.9% [12.0–57.0]; $p=0.032$).

To establish whether the IFN-I signature was modulated after treatment with ustekinumab, IFN-I signature scores were generated using a published composite 21-gene IFN-I gene signature.³³ IFN-I signature scores were increased in two-thirds of patients at baseline versus the 31 healthy control samples and were generally stable from weeks 0 to 24 in both treatment groups, suggesting that treatment with ustekinumab (or placebo) does not modulate this pathway (appendix). No significant differences were found when comparing expression of the IFN-I signature between baseline and longitudinal assessments in patients in the ustekinumab group. Thus, ustekinumab-mediated disease activity improvements were not associated with significant changes in IFN-I signature expressions after treatment. Although the proportions of patients achieving an SRI-4 response at week 24 were greater in the low IFN-I subset (18 [82%] of 22 patients in the ustekinumab group and six [55%] of 11 patients in the placebo group) than in the high IFN-I subset (19 [51%] of 37 patients and seven [23%] of 30 patients), the treatment effect size was similar in both subsets (low IFN-I difference of 27% between treatment groups and high IFN-I difference of 28%; figure 3). Together, these data suggest that ustekinumab does not modulate the IFN-I pathway and that patients with both high and low IFN-I signatures at baseline might respond to ustekinumab.

Between weeks 0 and 24, 47 (78%) of 60 patients in the ustekinumab group and 28 (67%) of 42 patients in the placebo group had at least one adverse event (table 3). Infections were the most common type of adverse event (27 [45%] in the ustekinumab group vs 21 [50%] in the placebo group). The most common adverse events were urinary tract infection, nasopharyngitis, upper respiratory tract infection, and headache (table 3). Among patients in the ustekinumab group, one patient had an infusion

reaction during the first administration (week 0) and subsequently discontinued study treatment. This was reported as a suspected anaphylactic reaction, characterised by dyspnoea and tachypnoea on physical examination, but without any fever, hypotension, involvement of the skin or mucosal tissue, or gastrointestinal symptoms. One patient had mild injection site reactions (urticaria) that did not change in severity with continued study treatment.

No deaths occurred between weeks 0 and 24. Nine patients had at least one serious adverse event up to week 24: five (8%) of 60 patients in the ustekinumab group (acute pancreatitis, bronchitis, pneumonia, anaphylactic reaction, and ischaemic stroke) and four (10%) of 42 patients in the placebo group (gastric ulcer, pyrexia, keratoacanthoma, and acute kidney injury). The ischaemic stroke event occurred in a 56-year-old woman who had multiple risk factors at baseline (long-standing hypertension requiring use of multiple antihypertensive drugs, presence of lupus anticoagulant, tobacco use, and chronic glucocorticoid use). A serious infection (the bronchitis and pneumonia previously mentioned) was reported in two patients; no cases of tuberculosis were reported. No treatment-emergent opportunistic infections, cases of herpes zoster, or malignancies occurred between weeks 0 and 24.

Steady state was achieved at about 15–17 weeks after the first administration of ustekinumab, and the median trough level at week 24 was 2.51 µg/mL. Between weeks 0 and 24, 59 patients in the ustekinumab group received at least one dose of ustekinumab and had at least one post-baseline serum sample taken. Of these, five (9%) patients tested positive for antibodies to ustekinumab, with two (3%) patients testing positive for neutralising antibodies.

Discussion

To the best of our knowledge, this study is the first to report efficacy and safety of an anti-IL-12 and IL-23 antibody in systemic lupus erythematosus. In this randomised controlled trial, a significantly greater proportion of ustekinumab-treated patients achieved the SRI-4 primary endpoint at week 24 than those who received placebo, when added to standard-of-care therapy. Similarly, greater proportions of ustekinumab-treated patients achieved more stringent levels of global response (SRI-5 or SRI-6) at week 24 than those who received placebo. Additionally, treatment benefit with ustekinumab was observed across several endpoints assessing joint and mucocutaneous disease and clinical biomarkers of lupus disease activity, including anti-dsDNA and C3. Furthermore, patients in the ustekinumab group were significantly less likely to have a flare than those in the placebo group.

Despite clear responses in the primary outcome and several other outcomes, the BICLA response was not statistically different between the groups. We performed a detailed review of SRI-4 responders who did not achieve a BICLA response. Several patients receiving ustekinumab

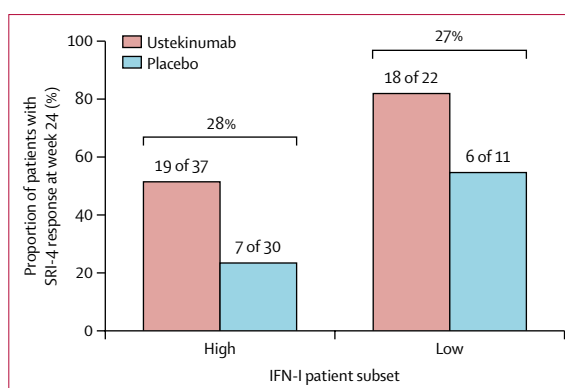


Figure 3: SRI-4 responses in high and low IFN-I patient subsets

On each bar, the number of patients achieving an SRI-4 response at week 24 is shown over the total number of patients in that subset at baseline. One patient in the ustekinumab group and one patient in the placebo group were not included in microarray analyses because of insufficient sample material or material that did not pass quality control parameters. Brackets show the percentage differences between the ustekinumab group and the placebo group in each patient subset. SRI-4=systemic lupus erythematosus disease activity index 2000 responder index-4. IFN-I=type I interferon.

	Ustekinumab (n=60)	Placebo (n=42)
Any adverse event	47 (78%)	28 (67%)
Infections	27 (45%)	21 (50%)
Most common adverse events*		
Urinary tract infection	6 (10%)	5 (12%)
Nasopharyngitis	6 (10%)	3 (7%)
Upper respiratory tract infection	5 (8%)	9 (21%)
Headache	4 (7%)	5 (12%)
Diarrhoea	4 (7%)	0
Worsening of systemic lupus erythematosus	3 (5%)	2 (5%)
Nausea	3 (5%)	2 (5%)
Bronchitis	3 (5%)	0
Pharyngitis	3 (5%)	0
Pharyngotonsillitis	3 (5%)	0
Pyrexia	3 (5%)	1 (2%)
Serious adverse event	5 (8%)	4 (10%)
Serious infections	2 (3%)	0

Data are n (%). No deaths or malignancies were reported in either group. Patients with more than one of the same event were counted once for that event. *Events that occurred in at least 5% of patients in the ustekinumab group.

Table 3: Patients with adverse events up to week 24

showed improvements, but not across all the BILAG domains that scored moderate to severe at baseline, which is required to achieve a BICLA response. Notably, the BICLA response is based on the BILAG-2004, which measures incremental disease improvement or worsening, whereas the SRI-4 response uses SLEDAI-2K, a weighted scale that measures whether clinical signs and symptoms are present or absent. Although the SRI-4 and BICLA responses were inconsistent in this study, the totality of the efficacy findings suggests that ustekinumab

might benefit both global and organ-specific disease activity in systemic lupus erythematosus.

Adverse events between weeks 0 and 24 of this study were similar to those seen in previous trials of ustekinumab in patients with psoriasis, psoriatic arthritis, and Crohn's disease.²⁶ Although the phase 2 clinical trial reported here is of modest size, the safety profile of ustekinumab in patients with other immune-mediated diseases has been established in multiple large phase 3 clinical trial programmes²⁶ as well as in extensive post-marketing surveillance (estimated cumulative worldwide exposure to ustekinumab from launch to Dec 31, 2017, is 1036701 person-years [data on file, Janssen Research & Development, LLC]). Notably, the ustekinumab dosing regimen used in this trial used a single weight-based intravenous administration followed by subcutaneous maintenance therapy. This dosing regimen is approved for the treatment of patients with Crohn's disease and will be further assessed in a phase 3 study of patients with systemic lupus erythematosus (ClinicalTrials.gov, number NCT03517722).

Given the heterogeneous presentation of systemic lupus erythematosus signs and symptoms, measuring changes in disease activity and response to therapy has been challenging. The SRI-4 response was developed to assess improvement in global disease activity and is based on at least a 4-point improvement in the SLEDAI-2K score without worsening in other global measures of disease activity (BILAG-2004 or PGA). Although the SRI response is not commonly used in clinical practice, this endpoint has been the basis for approval by regulatory agencies when assessing efficacy in patients with systemic lupus erythematosus.³¹ Achieving an SRI-4 response has been associated with improvements in assessments that are not included in the SRI criteria, such as CLASI and swollen and tender joint counts.³⁴ Prevention of flares is another crucial treatment goal in patients with systemic lupus erythematosus, because flares are associated with organ damage, morbidity, and mortality.³⁵ Thus, the findings in this trial suggest clinically meaningful effects of ustekinumab on systemic lupus erythematosus disease activity and flares.

The clinical efficacy in this study implicates IL-12 or IL-23, or both, as important cytokines in the immunopathology of systemic lupus erythematosus. Concentrations of both Th1-type cytokines and IL-12 are increased in patients with lupus.^{21,36} IL-12 is important in T-cell differentiation to Th1 and T follicular helper cells as well as the activation and function of cytotoxic T cells. Thus, attenuation of this pathway might benefit patients with systemic lupus erythematosus. The IL-23 and IL-17 axis has emerged as a major immunological pathway implicated in several autoimmune diseases, and increasing evidence suggests the involvement of this pathway in systemic lupus erythematosus as well.²⁴ Thus, experimental data indicate a role for both

IL-12 and IL-23 in the immunopathology of systemic lupus erythematosus. Further clinical, in-vitro, and ex-vivo studies will be needed to identify conclusively which cytokines are most important in this complex disease.

The proportion of patients with the IFN-I signature has been reported to range from about 50% in adult patients with systemic lupus erythematosus to nearly all paediatric patients.¹⁹ A 2017 clinical trial of a drug targeting the IFN-I receptor showed a treatment effect beyond placebo only in patients with an increased IFN-I signature.¹⁷ In our trial, the absence of modulation of the IFN-I pathway and the observed clinical response suggests that patients with systemic lupus erythematosus might respond to ustekinumab regardless of baseline IFN-I signature status, in keeping with the unique mechanism of action of this drug.

As expected in a phase 2 study, this trial is limited by the patient population size compared with phase 3 trials. However, the study was adequately powered to detect a difference between the treatment groups and assessed several clinically meaningful endpoints. Larger randomised, placebo-controlled, phase 3 trials are needed to confirm our findings. The pharmacokinetics, immunogenicity, and safety profile of ustekinumab in patients with systemic lupus erythematosus up to week 24 were consistent with previous trials of ustekinumab in other immune-mediated diseases.²⁶ In summary, the results of this study confirm the importance of IL-12 and IL-23 in this disease, and indicate that ustekinumab might be an effective therapy for systemic lupus erythematosus.

Contributors

RFvV, BHH, CLW, PL, ZT, RMG, BZ, BH, MC, MT, JLJ, and SR designed the study. RFvV, BHH, GCT, CLW, PL, ZT, VPW, RMG, BZ, BH, MC, MT, JLJ, and SR did acquisition, analysis, and interpretation of the data. All authors drafted or revised the manuscript with the assistance of a professional medical writer employed by Janssen Scientific Affairs, LLC.

Declaration of interests

RFvV reports grants from AbbVie, ArthroGen, Bristol-Myers Squibb, GlaxoSmithKline (GSK), Lilly, Pfizer, and UCB, and personal fees from AbbVie, AstraZeneca, Biotest, Bristol-Myers Squibb, Celgene, GSK, Janssen, Lilly, Medac, Merck, Novartis, Pfizer, Roche, and UCB. BHH reports grants and personal fees from Bristol-Myers Squibb and personal fees from Celgene and Janssen. GCT reports consulting fees from Janssen and advisory board fees from Abpro and Silicon Therapeutics. PL and ZT report consulting fees from Janssen. VPW reports grants and personal fees from Janssen. MT is an employee of Janssen Biologics BV. CLW, RMG, BZ, BH, MC, JLJ, and SR are or were employees of Janssen Research & Development, LLC, and own stock in Johnson & Johnson, of which Janssen Research & Development, LLC, is a wholly owned subsidiary.

Data sharing

The data sharing policy of Janssen Pharmaceutical Companies of Johnson & Johnson is available online. As noted on the website, requests for access to the study data can be submitted through the Yale Open Data Access Project website.

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